Universally Legal Response Pairings in Gravity Connect Four

Classification, obstructions, and certified midgame bounds

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Developmental research note and proof audit

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Abstract

We formalize a restricted class of fixed response strategies for finite gravity Connect Four and prove a structural classification by isolated column advancement. On an odd-width, odd-height board after one opening disc, this classification yields an exact counting formula. We then prove that on every nine-column board of odd height at least five, every universally legal fixed response matching for the first player admits a second-player counterattack within sixteen plies. Separately, we give a sufficient, certificate-producing midgame theorem based on vertical pairs and currently playable cross-pairs. An explicit reachable draw and an eight-node winning endgame certificate illustrate the distinction between a position proof and an opening solution. The associated computation completed a 27-ply finite-horizon search of the empty 9×9 board, but did not determine its full-game value. The present note therefore does not claim a weak or ultra-weak solution of that board, or priority for the underlying pairing ideas.

Status. The value of the empty 9×9 board remains unknown in this research record. Theorems below concern stated strategy classes and sufficient position certificates. Search statistics are reported separately as computational observations.

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1 Game, notation, and the unresolved objective

Let

$$\mathcal{B}_{w,h} = \{(c, r) : 1 \le c \le w,\ 1 \le r \le h\},$$

with row 1 at the bottom. A gravity position has occupied cells $\{(c, 1), \dots, (c, a_c)\}$ in column c, for heights $0 \le a_c \le h$. A move in a nonfull column c occupies $(c, a_c + 1)$. The players alternate, with P_1 first. A player wins immediately on obtaining four of its discs in a horizontal, vertical, or diagonal line. Play ends at the first win, or with a draw when all cells are occupied without a win. There are no passes, removals, wraparound columns, or misère rules.

Let $\mathcal{W}_{w,h}$ be the family of four-cell winning sets. For $w, h \ge 4$,

$$|\mathcal{W}_{w,h}| = h(w - 3) + w(h - 3) + 2(w - 3)(h - 3). \tag{1}$$

Thus $|\mathcal{W}_{9,9}| = 54 + 54 + 36 + 36 = 180$. Equation (1) follows by choosing the leftmost cell of a horizontal four, the bottom cell of a vertical four, and the bottom endpoint of either diagonal orientation.

For a legal nonterminal position S, define $\mathrm{V}(S) \in \{-1, 0, 1\}$ from the perspective of the player to move: loss, draw, and win, respectively, under optimal play. At a terminal position immediately after an opponent's win, set $\mathrm{V} = -1$; at a terminal draw, set $\mathrm{V} = 0$. Backward induction gives

$$\mathrm{V}(S) = \max_{c \in L(S)} \{-\mathrm{V}(S_c)\}, \tag{2}$$

where $L(S)$ is the set of nonfull columns and S_c is the position after playing c. This is a finite recursion because each move consumes a cell.

Definition 1.1. A *weak solution* determines the empty-board value and supplies a strategy attaining that value against every legal opposing continuation. An *ultra-weak solution*, in the terminology used here, establishes the empty-board value without necessarily constructing an executable strategy.

Neither objective follows from a favorable principal variation, an unbeaten self-play record, a nonzero heuristic score, or a timeout. In particular, our restricted obstruction below is not a second-player solution of the full game. The strategic-rule tradition for Connect Four includes Allis's Claimeven and Baseinverse ideas [1]; our pairing formalization is presented as a self-contained research development, without a literature priority claim.

2 Universally legal fixed response matchings

The classification concerns gravity and occupancy only. For its definition, we temporarily ignore the rule stopping play after four in a row.

Definition 2.1. Fix an initial gravity occupancy S with an even number of empty cells $\mathcal{E}(S)$. A *response matching* M is a perfect matching of $\mathcal{E}(S)$. An initiator chooses any legal empty cell x; the responder must immediately occupy its mate $M(x)$. A *completed-round state* is a state reached from S after finitely many such legal two-cell rounds.

The matching is *universally legal*, abbreviated UL, if at every completed-round state, every legal initiating move has an immediately legal mate response. This requirement is imposed even on occupancies that would already have ended a Connect Four game.

This last qualification is substantial. A matching with an illegal response only after the responder has already won is not necessarily UL. Adaptive matchings and fallback moves are also outside this definition.

A *front* of a nonfull column is its lowest empty cell. For a column with $r_i = h - a_i$ empty cells, enumerate its empty cells from bottom to top as $u_{i,1}, \ldots, u_{i,r_i}$.

Lemma 2.2 (Initial fronts)**.** *At the start of a universally legal response process, the mate of each front is either the cell immediately above it in the same column, or the front of a different column.*

Proof. Let x be a front. Before the initiating move at x, the only legal cells are the current fronts. After that move, the only newly legal cell is the cell immediately above x, if it exists. Its mate must be one of these legal cells. A different cell higher in another column remains unsupported. □

Lemma 2.3 (Isolated advancement)**.** *Suppose that at a completed-round state the initial fronts of a collection of columns are matched internally in that collection, and all other columns are kept frozen.*

(i) If a column's front is matched vertically to the next cell, all its remaining cells are paired into consecutive vertical dominoes.

(ii) If the fronts of two columns are matched to each other, their cross-column pairs form a common initial prefix of their remaining cells. After the prefix ends, all remaining pairs in each column are consecutive vertical dominoes.

Proof. For (i), consume the front's vertical pair while leaving every other column unchanged. By lemma 2.2, each frozen front was initially matched either to another initial front or to its own immediate successor. In particular, none was matched to a cell newly exposed in the advancing column. A cell higher in a frozen column is not playable. By universal legality, the newly exposed front must therefore be paired vertically to the cell directly above it. Repeating the same argument advances this column alone to completion. Its remaining cell count must be even.

For (ii), consume the cross-pair and keep all other columns frozen. The new fronts, when both exist, cannot be matched to any outside front: those fronts have their own fixed mates. They cannot be matched to any higher outside cell, because it is unsupported. By lemma 2.2, either the two new fronts are paired to each other, or at least one of them is paired vertically.

If one front is paired vertically, the other cannot be paired to either cell of that vertical pair, since both mates are already fixed. It also cannot be paired to a higher cell in that column: that cell is not playable at the present state, whereas the other front can be initiated immediately. It must therefore be paired vertically as well. Part (i), now advancing either column with every other column frozen, forces vertical pairs above that point in both columns. If instead a column is exhausted at the end of a cross-pair, the other column, if nonempty, is forced to finish vertically by the same isolated-advancement argument. Consequently the cross-pairs form a prefix and cannot resume after a vertical split. □

Theorem 2.4 (Structure of universally legal response matchings)**.** *For an arbitrary initial gravity occupancy, a response matching is universally legal if and only if the nonempty columns admit a partition into components of the following two types:*

(a) a single column with an even number of empty cells, matched into consecutive vertical dominoes;

(b) two columns i, j with k cross-pairs

$$\{u_{i,t}, u_{j,t}\}, \qquad 1 \le t \le k,$$

for some $1 \le k \le \min(r_i, r_j)$, followed by consecutive vertical dominoes in each column. Necessarily $r_i - k$ and $r_j - k$ are even.

The component partition and each nonempty cross-prefix are determined by the matching.

Proof. By lemma 2.2, the initial fronts are paired either vertically within their columns or reciprocally with another front. This partitions the nonempty columns initially into singletons and two-column components. No column belongs to two components, because a front has one mate.

Each component can be advanced on its own, leaving all other components frozen. The argument in lemma 2.3 shows that it cannot acquire a later connection to another component: an outside front already has its fixed mate, and every higher outside cell is unsupported when the component is advanced alone. The same lemma supplies exactly the vertical singleton form and the cross-prefix–then–vertical form. The parity conditions follow because the tails consist of dominoes. Initial front mates determine the components; the maximal consecutive cross-prefix determines k.

Conversely, consider a matching of the asserted form. In a singleton, every initiating move at a domino's lower cell exposes its upper mate. In a two-column component, each cross-pair consumes one cell in each column, keeping the two fronts synchronized by relative depth until the prefix ends. Both front cells of the next cross-pair are then playable. Once the prefix is exhausted, the columns are independent vertical stacks of dominoes. Interleaving rounds from different components changes none of these facts. Every prescribed response is therefore legal. □

2.1 Counting after one opening disc

Corollary 2.5. *Let $w = 2m + 1$ and $h = 2n + 1$. After one bottom-row opening disc in a fixed column q, the number of universally legal response matchings is*

$$N_{w,h} = (2m - 1)!!\,(n + 1)^m, \tag{3}$$

with $(-1)!! = 1$ when $m = 0$.

Proof. The opening column has $2n$ empty cells; every other column has $2n + 1$. Two columns in a cross component have the same empty-count parity by theorem 2.4. Thus the unique even column cannot join an odd column and is a vertical singleton. Each odd column must join another odd column. There are $(2m - 1)!!$ perfect matchings of these $2m$ columns. Each paired component permits exactly the odd prefix lengths $1, 3, \ldots, 2n + 1$, giving $n + 1$ choices independently for each of the m components. The representation is unique, so the product neither omits nor double-counts a matching. □

For 9×9, this is

$$N_{9,9} = 7!!\, 5^4 = 105 \cdot 625 = 65\,625$$

for each fixed opening, or 590 625 configurations over all nine openings. Reflection allows the five representative openings $q = 1, \ldots, 5$ to cover the remaining four. The enumeration therefore checked $5N_{9,9} = 328\,125$ configurations. This last number is *not* the number of distinct matching orbits under reflection; at the central opening, reflection can still identify two of the checked matchings.

3 A counterattack theorem for nine columns

The next theorem is an extension of the recorded 9×9 obstruction. Its proof uses only that the height is odd and at least five.

Theorem 3.1 (Restricted response obstruction)**.** *On a $9 \times H$ board with odd $H \geq 5$, let P_1 open in any column and thereafter respond according to a fixed universally legal perfect matching of the remaining cells. Then P_2 has a counterstrategy that wins by game ply 16, or earlier. Hence no strategy in this specified family guarantees P_1 a nonloss.*

Proof. By theorem 2.4 and corollary 2.5, the opening column is vertical, and the other eight columns form four paired components with odd cross-prefix lengths. Write q for the opening column.

Case 1: some cross-prefix has length at least three. Choose one column a in that component, whose partner is b. Player P_2 drops its first four discs consecutively in column a. The first three prescribed replies lie in column b, at rows $1, 2, 3$. Player P_2 therefore wins vertically at $(a, 4)$, on game ply 8. The cell exists since $H \geq 5$. Before that move, P_1 has only the opening disc in column $q \notin \{a, b\}$ and at most three discs in column b. Such a set cannot contain four collinear cells: a vertical line in b has at most three, and a nonvertical line contains at most one cell of each column. Thus P_1 cannot have won earlier.

Case 2: every cross-prefix has length one. Only the bottom cells of each paired component are cross-paired; above them, all columns have vertical pairs $(2, 3), (4, 5), \ldots$. For each of the four bottom pairs, let P_2 choose one endpoint and let P_1 receive the other as its prescribed response. We first show that these choices can avoid a P_1 bottom-row four.

Choose the four orientations uniformly and independently. An interval of four bottom cells containing q is wholly P_1 with probability at most 2^{-3}: if two of its other cells are paired to each other the probability is zero; otherwise it fixes three independent orientations. An interval not containing q has probability at most 2^{-4} by the same reasoning. There are six four-cell intervals, at most four containing q. The union bound gives

$$\Pr(\mathsf{P}_1 \text{ has a bottom-row four}) \leq \frac{4}{8} + \frac{2}{16} = \frac{5}{8} < 1. \tag{4}$$

Consequently a safe orientation exists.

Fix such an orientation and fill the four bottom pairs, in any order. Player P_1 cannot win during this preparation: its occupied cells are a subset of its eventual safe bottom-row set. If P_2 wins during preparation, the theorem already holds. Otherwise the entire bottom row is now full, after game ply 9.

Player P_2 next plays row 2 in columns $1, 2, 3, 4$ in that order. Each prescribed reply is row 3 in the same column. The fourth initiating move completes a horizontal four for P_2, on ply 16. Before that

move, P_1 has no row-2 cells, no cells above row 3, and at most three row-3 cells. Its bottom row has no four by construction. It cannot therefore win horizontally, vertically, or diagonally before P_2 wins. This completes both cases. □

Corollary 3.2 (Exact 9×9 enumeration split)**.** *For five representative openings,* 327 600 *classified configurations have a prefix of length at least three. The remaining* 525 *have every prefix length one. Their respective counterattacks terminate by plies* 8 *and* 16*.*

Proof. For each fixed opening, the all-one-prefix choices number 105; all other choices number $65\,625 - 105 = 65\,520$. Multiply each number by five. □

Remark 3.3 (Quantifier scope)*.* The conclusion is

$$\forall M \in \mathrm{UL} \quad \exists \text{ a } \mathsf{P}_2 \text{ counterstrategy against } M.$$

It is not a statement that P_2 beats arbitrary P_1 strategies, and it does not provide a universal counterstrategy against a hidden or adaptively changing matching. The full-game value is not determined.

4 A sufficient midgame certificate

Let C be the player to move and D the responder in an ongoing position S, with occupied sets A_C, A_D. Suppose $|\mathcal{E}(S)|$ is even. In every column with an even number of empty cells, pair successive empty cells vertically. In each column with an odd number, reserve its lowest empty cell and vertically pair the cells above it. The number of reserved cells is even, so match them in arbitrary cross-column pairs.

Denote the vertical pairs by M_v, the reserved-cell pairs by M_b, and write

$$U = \{\text{upper endpoint of a pair in } M_v\}, \qquad G = A_D \cup U.$$

The reserved cells can have different absolute row coordinates; all are currently playable, which is the fact needed for legality.

Lemma 4.1. *The strategy in which D completes the pair containing C's latest move is universally legal. Player C never obtains a cell of U, and obtains at most one endpoint of each pair in M_b.*

Proof. The reserved bottom cells of an unconsumed cross-pair are both playable. Consuming the pair leaves even empty heights in its two columns. All other rounds consume a consecutive vertical domino; only its lower cell is playable before the round, and the initiating move exposes the upper response. An induction on completed rounds proves legality and assigns each vertical upper cell to D. Pair completion gives opposite owners to the two cells of every cross-pair. □

Theorem 4.2 (Line-cover certificate)**.** *With the construction above, suppose*

$$\forall L \in \mathcal{W}_{w,h}, \qquad L \cap G \neq \varnothing \quad \textit{or} \quad \exists \{u, v\} \in M_b : \ \{u, v\} \subseteq L. \tag{5}$$

Then $\mathrm{V}(S) \leq 0$*. If, in addition,*

$$\exists L_D \in \mathcal{W}_{w,h} \quad L_D \subseteq G, \tag{6}$$

then $\mathrm{V}(S) = -1$*.*

Proof. Fix any winning line L for the current player C. If it meets A_D, that opposing disc blocks it permanently. If it meets U, the cell is never owned by C by lemma 4.1; before it is filled, its being empty already prevents completion by C. Otherwise (5) supplies a cross-pair inside L, and C cannot obtain both endpoints. Thus no legal initiating continuation can give C a win. This proves $\mathrm{V}(S) \le 0$.

If (6) holds, continue the response strategy. The board is finite and each completed round consumes two cells. Unless D has already won, play must reach full occupancy, at which point all cells of U have been obtained by D. Then L_D is a completed D line, contradicting a terminal draw. Since C cannot win, D forces a win and $\mathrm{V}(S) = -1$. □

4.1 A finite matching-cover search

Let O be the reserved cells and define the remaining constraints

$$\mathcal{R} = \{L \cap O : L \in \mathcal{W}_{w,h},\ L \cap G = \varnothing\}.$$

A certificate exists in this construction exactly when there is a perfect matching of O such that every $R \in \mathcal{R}$ contains one of its pairs. A constraint with $|R| < 2$ rules out this construction immediately. One can branch on a constrained R, select two of its still-unmatched vertices, remove constraints covered by that pair, and recurse.

This backtracking is complete for the stated construction: every full solution must select a pair in the chosen constraint, so one branch agrees with it. Once all constraints are covered, the unused reserved vertices have even cardinality and may be paired arbitrarily. On a nine-column board, $|O|$ is even and at most eight. Failure to find such a matching establishes only failure of this sufficient certificate format, not a winning value for C.

4.2 A fully explicit winning endgame

Use columns $A, \dots, I$ and write X for P_1, O for P_2. After the 74-ply history in appendix A, the board is

	A	B	C	D	E	F	G	H	I
9	X	$\cdot$	O	O	$\cdot$	X	$\cdot$	O	$\cdot$
8	O	O	X	O	$\cdot$	X	X	X	O
7	O	X	X	X	$\cdot$	O	O	O	X
6	X	O	X	O	$\cdot$	X	X	O	O
5	X	O	O	X	X	O	O	X	X
4	X	X	O	X	O	X	X	O	O
3	O	O	O	X	X	O	X	X	O
2	O	O	X	O	X	X	O	O	X
1	X	O	X	X	X	O	O	X	O

(7)

Its height vector is $(9, 8, 9, 9, 5, 9, 8, 9, 8)$ and it is P_1's turn.

Proposition 4.3. *In* (7)*,* P_1 *wins within three further plies by playing* $B9$ *and then responding in column* E*.*

Proof. The move $B9$ is legal. The only remaining nonfull columns are then E, G, I. If P_2 plays $E6$, the legal response $E7$ completes the horizontal line $B7, C7, D7, E7$. If P_2 instead plays $G9$

or $I9$, the legal response $E6$ completes the diagonal $B9, C8, D7, E6$. None of the three opposing moves is a win, as checked directly from (7). These are all legal replies, so the strategy covers every opponent continuation. The corresponding rooted certificate has one root, one node after $B9$, three opponent-move nodes, and three winning leaves: eight nodes in total. □

5 An explicit reachable draw

The existence of a legal drawn game is different from the existence of a strategy forcing a draw. It does, however, refute any argument that odd board area by itself makes a draw impossible.

Proposition 5.1. *There exists a legal drawn game of length* 81 *on the* 9×9 *board.*

Proof. Take the 81-digit history σ in appendix A. Each digit occurs nine times, so no move overfills a column. Alternating odd and even plies gives the following final board:

	A	B	C	D	E	F	G	H	I
9	X	O	X	O	O	X	O	O	O
8	X	O	O	O	X	O	O	O	X
7	O	X	X	X	O	X	X	O	O
6	X	X	O	X	X	X	O	X	X
5	O	X	X	O	O	O	X	O	X
4	X	O	O	O	X	X	X	O	X
3	O	X	X	X	O	X	O	X	O
2	X	X	O	O	X	X	O	X	O
1	X	O	X	O	X	O	X	O	O

It contains 41 discs of X and 40 of O. For an exact arithmetic check of all winning lines, index a cell by $j = 9(r-1) + (c-1)$ and encode a color set as an integer whose jth bit is its occupancy. In hexadecimal,

$$X = \texttt{25889bb7746b8ab96655},$$
$$O = \texttt{1da7764488b94754699aa},$$
$$X \mid O = 2^{81} - 1 = \texttt{1ffffffffffffffffffff}.$$

For a positive direction shift d, let Q_d mark all valid starting cells of a four in that direction. The exact masks are

d	direction	Q_d (hexadecimal)
1	horizontal	`3f1f8fc7e3f1f8fc7e3f`
9	vertical	`3fffffffffffff`
10	rising right	`7e3f1f8fc7e3f`
8	rising left	`3f1f8fc7e3f1f8`

Equivalently, the masks have the exact formulas

$$Q_1 = 63 \sum_{r=0}^{8} 2^{9r}, \quad Q_9 = 2^{54} - 1,$$
$$Q_{10} = 63 \sum_{r=0}^{5} 2^{9r}, \quad Q_8 = 504 \sum_{r=0}^{5} 2^{9r}.$$

Direct integer calculation gives, for each $d \in \{1, 9, 10, 8\}$ and each $Z \in \{X, O\}$,

$$Q_d \,\&\, Z \,\&\, (Z \gg d) \,\&\, (Z \gg 2d) \,\&\, (Z \gg 3d) = 0. \tag{8}$$

Here & and $\gg$ denote bitwise AND and right shift. The starting masks exclude row wraparound, so (8) checks exactly the 180 winning sets. Neither color has four on the final board. Since a color's occupied set only grows, no earlier prefix could have contained four either. The sequence is therefore legal up to its full-board draw. □

6 What finite-horizon computation can establish

Define $\mathrm{V}_d(S)$ by using the same terminal values as V and the recursion (2), but assigning value zero to every nonterminal position at depth zero. A depth-zero cutoff is an unresolved leaf, not an assertion that the full game is drawn.

Proposition 6.1. *If an exact finite-horizon computation gives $\mathrm{V}_d(S) = 1$, then $\mathrm{V}(S) = 1$. If it gives $\mathrm{V}_d(S) = -1$, then $\mathrm{V}(S) = -1$. If $\mathrm{V}_d(S) = 0$, neither player has a strategy forcing its own win within d additional plies, but the full-game value can still be any of $-1, 0, 1$.*

Proof. The nonzero implications follow by induction on d. A positive value has a child of negative value; by induction the opponent loses in that child. A negative value has every child positive; by induction the opponent wins after every legal choice. The terminal cases are immediate. Equivalently, in the truncated game a nonzero value is certified by a strategy subtree all of whose relevant leaves are actual terminal wins, since cutoff leaves have value zero.

A zero value is the minimax value of a finite game in which unresolved cutoffs pay zero. Backward induction gives each player a strategy preventing the other from obtaining a positive terminal payoff before the cutoff. It does not determine what occurs beyond the cutoff. □

Recorded computational observation. The corrected implementation reported

$$\mathrm{V}_{27}(S_\varnothing) = 0.$$

This is a logged software result, not a separate machine-checked formal proof object. The depth-28 iteration did not finish. The final record contains 726 138 880 node visits including that unfinished iteration; the completed depth-27 record contains 338 098 896 cumulative visits. Tests compare 1,000 short-horizon cases with a plain-array implementation and 100 endgames with independent full-depth values. These tests increase confidence in the implementation but do not constitute an opening solution.

7 Certificate and search semantics

7.1 Strategy-tree certificates

For a claimed winner, a strategy-tree certificate must retain at least one legal prescribed move on that player's turns and *every* legal move on the opponent's turns. Every leaf must be an actual win for the claimed winner. A verifier replays each edge under the rules and rejects omitted opponent replies, illegal moves, and false terminal leaves.

Proposition 7.1. *A finite strategy tree satisfying those conditions proves a forced win at its root.*

Proof. Induct upward from winning leaves. At the claimed winner's nodes, a retained child supplies a winning move. At the opponent's nodes, every legal move is retained and leads to a winning child. The root is therefore won regardless of the opponent's choices. □

7.2 Proof-number search and WDL bounds

For the Boolean proposition "P_1 can force a win," P_1 moves are OR nodes and P_2 moves are AND nodes. At a true terminal proposition assign proof/disproof numbers $(0, \infty)$; at a false one assign $(\infty, 0)$. An unresolved frontier starts with $(1, 1)$. The recurrences are

$$(p, d)_{\mathrm{OR}} = \left(\min_i p_i,\ \sum_i d_i \right),$$

$$(p, d)_{\mathrm{AND}} = \left(\sum_i p_i,\ \min_i d_i \right).$$

A zero proof number is a true proposition by the same OR/AND induction as proposition 7.1; a zero disproof number is a false proposition. Positive numbers are search bookkeeping, not probabilities or percentages complete. The recorded depth-first proof-number searches timed out with both root numbers positive.

A WDL cache stores exact values or inequalities $\mathrm{V}(S) \geq a$ and $\mathrm{V}(S) \leq b$. Its Boolean interpretation must account for whose turn it is. For example, $\mathrm{V}(S) \leq 0$ disproves a P_1 win when P_1 is to move, whereas $\mathrm{V}(S) \geq 0$ disproves it when P_2 is to move. A draw is also a false instance of "P_1 can force a win." The implementation checks all 18 combinations of turn parity, bound type, and bound value against the allowed outcomes.

7.3 Hashing and compatible cached bounds

The main board representation uses nine columns of ten bits, including a header bit per column, so a position key needs up to 90 bits and is stored in a 128-bit integer. A table bucket index is only a lookup aid. A retrieved entry is accepted only after comparison of the complete normalized key. Consequently an index collision can discard useful information but cannot by itself justify substituting a different position's value.

For several valid bounds on the same key, define

$$\ell = \max(\{-1\} \cup \{\text{stored lower bounds}\}), \qquad u = \min(\{1\} \cup \{\text{stored upper bounds}\}).$$

Necessarily $\ell \leq \mathrm{V}(S) \leq u$. Thus $\ell = u$ supplies an exact value, and $\ell > u$ detects incompatible claims. The final merger retained 16 650 distinct keys, with 11 051 exact values and no detected bound contradictions. Absence of contradictions is a consistency check, not an independent correctness proof of every cached value. The normalized-key and dead-stone reductions remain part of the trusted implementation.

8 Auditable evidence and research status

The mathematical statements in sections 2 to 4 are ordinary proofs under explicit hypotheses. The following finite checks are supporting evidence and reproducible artifacts:

- Exhaustive replay of all 328 125 classified configurations for five representative openings; the 525 all-one-prefix cases have explicit stored response matchings and counterattack histories.
- Independent paired-state legality enumeration of all 105 cell matchings in each of two 3×3 cases, and all 135 135 in each of one 3×5 and one 5×3 case. The accepted matchings agree exactly with theorem 2.4 in those cases.
- A self-contained manuscript audit also replays all 42 525 classified configurations at height five and all 134 400 at height seven, across five representative openings each. Every counterattack finishes by ply 16. These checks support the infinite-height-family argument in theorem 3.1; they do not replace its proof.
- Twenty-one independently constructed midgame line-cover certificates, including nine forced losses, on 100 independently solved endgames. Negative tests reject malformed or unjustified certificates.
- An eight-node endgame strategy certificate corresponding to proposition 4.3, and the full-board draw in proposition 5.1.
- Exact opening searches, horizon search, and proof-number searches whose unresolved outputs are preserved as unresolved.

The generalized exact engine is based on Steininger's published solver [2]; its research copy and license are retained. Tromp's reference page supplies historical context and board-size results [3]. None of the cited standard-board results is used to infer the 9×9 value. The elementary classification, its application, and the line-cover arguments are supplied in full here; this note is not a claim of peer review or proof-assistant verification.

8.1 Development directions with exact proof obligations

1. **Adaptive response systems.** Extend fixed matchings to states with finitely many response modes. A valid certificate must specify the mode transition after each legal opposing move and prove a preserved invariant. The obstruction in theorem 3.1 does not exclude such systems.
2. **Stronger strategic leaves.** Extend theorem 4.2 with other proved threat-control rules. An unverified heuristic must not be returned as an exact leaf value.
3. **Theorem-guided opening certificates.** Search for a finite opening tree closed under every opposing reply, whose leaves are actual wins or independently verified strategic certificates. By proposition 7.1, such a tree would prove an opening win.
4. **Ultra-weak arguments.** A nonconstructive argument must still establish the empty-board value. Odd area alone fails by proposition 5.1; failure of a restricted strategy class does not establish the opponent's full-game win. No such opening argument is claimed here.

A Exact move histories

Digits $1, \ldots, 9$ denote columns $A, \ldots, I$. Spaces below are grouping separators, not moves.

The 74-ply winning-endgame history.

```
394214915 666623782 294143441 343321958
378255783 311517697 648668187 724886992 79
```

Replaying it gives (7). No prefix has already won; the stored verifier checks each prefix before accepting the next move. The strategy is $B9$, followed by $E7$ after $E6$, or by $E6$ after $G9$ or $I9$.

The 81-ply drawn-game history.

```
391312592 458742532 275685278 998225892
717344155 499998768 414433753 463371118 176668666
```

Its column multiplicities are all nine and its final color sets are the integers in proposition 5.1. The bitwise identities (8) are an exact finite certificate of the absence of four.

B Reproduction manifest

Paths are relative to the research directory containing this LaTeX project. The source proofs and certificate checkers are intentionally separated from the large search programs.

- `pairing-refutations.py`: structural configurations and legal counterattack replay.
- `continuation/check-pairing-classification.py`: independent small-board enumeration of abstract matchings and paired-state legality.
- `pairing-proof.py`: independent line-cover certificate constructor and verifier; uses the standalone rules in `certificate.py`.
- `certificate.py`: exact endgame strategy-tree verification.
- `horizon.cpp` and `validate-horizon.cpp`: finite-horizon minimax and independent regression cases.
- `dfpn.cpp`: proof-number search, including seed-bound conversion.
- `merge-bounds.py`: duplicate-key grouping, bound compatibility, and exact-value extraction.
- `continuation/results.json`: frozen numerical record and source hashes. `continuation/REPORT.md` records computational context.

Representative commands, from that research directory, are:

```
python3 pairing-refutations.py
python3 continuation/check-pairing-classification.py
python3 pairing-proof.py
python3 certificate.py endgame-certificate.json
sh build.sh
./validate-horizon
./dfpn --validate continuation/merged-bounds.csv
```

The script `latex-development-note/audit_note.py` independently checks the manuscript's draw integers, the three-ply endgame, and the height-five and height-seven counterattacks, using only Python's standard library.

The associated files are development artifacts: future runs may produce new numerical records. This manuscript records the state reviewed on the date shown on its title page.

References

[1] V. Allis. *A Knowledge-based Approach of Connect-Four: The Game is Solved, White Wins.* Master's thesis, Vrije Universiteit Amsterdam, October 1988. `https://tromp.github.io/c4/connect4_thesis.pdf`.

[2] C. Steininger. *c4: A fast Connect 4 solver*. Source repository, research baseline commit `fa27f186d2f1bf8e8fd61bfad63454c6e4b0431c`. `https://github.com/ChristopheSteininger/c4`.

[3] J. Tromp. *John's Connect Four Playground*. `https://tromp.github.io/c4/c4.html`. Accessed 6 November 2026.